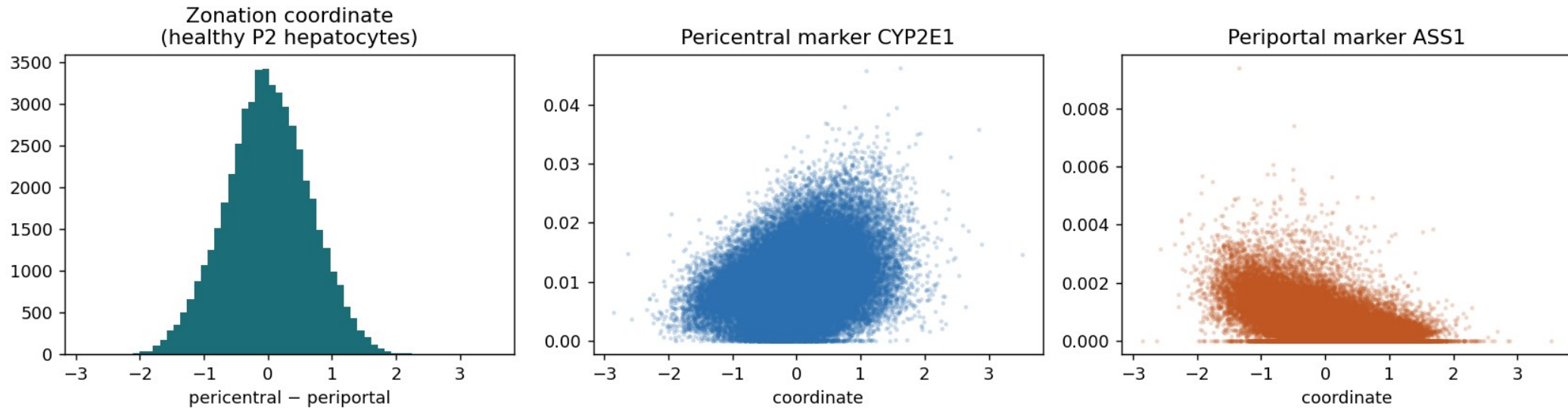


# Controls — does the result hold up?



✓ **Positive control** — on healthy liver the coordinate recovers zonation: pericentral CYP2E1 rises with it, periportal ASS1 falls (real data, above).

✗ **Negative control** — shuffle the disease-stage labels and re-run: the collapse trend must VANISH. A 'signal' that survives shuffling is an artefact, not biology.

# Limitations

**Ruler-validity — the real one.** Disease may corrupt the very marker genes the coordinate relies on, so a 'collapse' could be the ruler falling apart rather than true de-zonation. Step 5b diagnostics (internal coherence, PC-PP anti-correlation, split-half reproducibility) tell these apart before we make any claim.

**Few patients at the extremes.** The cohort thins at the ends — cirrhosis ( $n = 4$ ) and end-stage ( $n = 5$ ) — so the donor-level trend rests on a handful of people there; the tails of the curve carry the widest confidence intervals.

**Cross-dataset transfer.** The ruler is learned on a different healthy cohort and platform; residual batch effects could tilt the coordinate. We mitigate with rank/z features, a same-platform (snRNA) classifier, and the healthy positive control — but it remains a caveat.

**Minor caveat — spatial scrambling.** If cells kept their expression identity but were physically rearranged, dissociated data couldn't see it. We judge this biologically very unlikely in MASLD, and note it only for completeness.